Yipeng Li

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EDUCATION

Washington University in St. Louis **St. Louis, MO**
M. S in Computer Science ***Expected: May 2026***
N.

SKILLS

Programming Languages: Python, JavaScript/TypeScript, SQL, C++
Framework / Tools: Django, Flask, Node.js, Springboot, AWS (Lambda, EC2, DynamoDB), Flutter, Mockito
Database: MySQL / MariaDB, Redis

PROFESSIONAL EXPERIENCE

VisionX **San Jose, CA**
Software Engineer Intern **May 2025 - July 2025**

- Contributed to the development of RingX cross-platform mobile app supporting blood pressure, SpO_2 , heart rate, and pace tracking for 5,000+ elderly users, enabling caregivers to monitor health data in real time.
- Expanded and optimized a relational **SQL** database by designing and integrating new tables (SpO_2 , heart rate, sleep), normalizing schema, and adding indexes, reducing overall query latency by 30%.
- Implemented back-end APIs with **Django**, including: Real-time SpO_2 alert API delivering threshold warnings to caregivers; Analytics API generating weekly blood pressure and heart rate reports.
- Built cross-platform UI pages with **Flutter** (login, feedback, charts), ensuring a consistent and responsive experience across iOS and Android.
- Integrated **Apple HealthKit API** to retrieve step count, heart rate, and sleep data, synchronizing with the internal database to provide comprehensive health insights.

Tencent **Shenzhen, China**
Software Engineer Intern **May 2023 - Aug 2023**

- Developed and optimized an internal document management system for secure and efficient file transfers, including PDFs, spreadsheets, datasets, and images.
- Enhanced document security by implementing secure HTTPS communication protocols using **Flask** (**Python**), ensuring encrypted data transfer and secure interactions. Utilized parameterized queries to securely interact with MySQL, preventing SQL injection attacks. Additionally, implemented secure file storage solutions on **Tencent Cloud (CVM/COS)**, significantly improving reliability and confidentiality of internal documentation.
- Implemented LSTM-based (**Long Short-Term Memory**) machine learning recommendations to personalize onboarding experiences and boost user engagement.

Alibaba Group **Hangzhou, China**
Software Engineer Intern **Feb 2023 - Mar 2023**

- Developed and maintained an internal company shopping website for commemorative items and custom-designed products, providing service to approximately 500 internal users.
- Designed, implemented, and optimized database schemas using **MySQL** and integrated coupon management API via **Java Spring Boot**, to efficiently support coupon features, enabling employee discount programs.
- Implemented image caching, storing product images in **cache** upon user load to improve performance and enhance user experience.
- Conducted rigorous testing with **Mockito**, achieving 80% test coverage and ensuring system reliability.

PROJECT

AWS Serverless Web Application **Mar 2026**

- Built a **serverless** web application on AWS with two core modules: a Num2Words API with **DynamoDB**-backed history storage and a movie search system powered by Amazon OpenSearch Service.
- Implemented a Python-based Num2Words API with **AWS Lambda** and **API Gateway** to convert integer inputs into English words, persist request history in **DynamoDB**, and support interaction through a **React** frontend.
- Developed a full-stack movie search feature using **React**, **AWS Lambda**, **API Gateway**, and **OpenSearch**, enabling keyword-based search across movie titles and plot descriptions and returning structured JSON results for frontend rendering.
- Imported and indexed movie data in **OpenSearch**, validated search behavior with match and multi_match queries, and resolved dependency packaging, **IAM** authorization, OpenSearch role mapping, and timeout issues using **CloudWatch** and **AWS security** settings.